

ResQNet

Disaster preparedness and
awareness web app

MiniHackathon 2025

HackHers



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Problem Statement

Evacuations delayed
or not taken seriously.

Illnesses and unsafe
practices after disasters.

Vulnerable households
often miss timely aid.

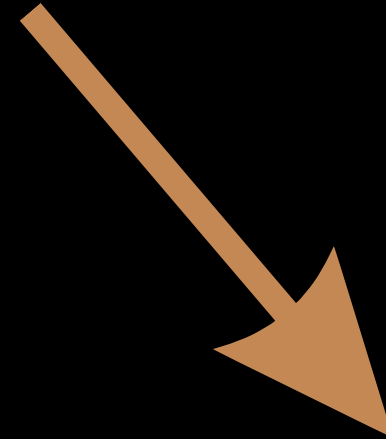
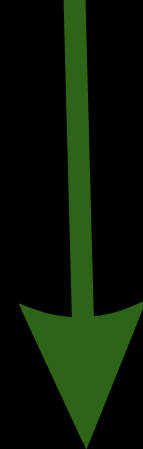
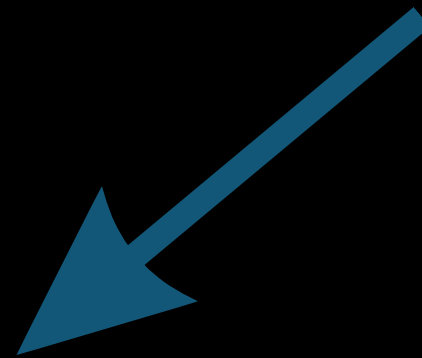
Chaos during floods/landslides
due to blocked routes

No simple, visual way
to understand risks.



Problem	Stat	Consequence
Evacuation Delays	6000+ evacuated late (Sri Lanka , 2024 floods)	More Exposure to floodwater, property damage, risk to lives
Blocked Roads	Roads inaccessible (Malaysia , Pakistan floods)	Relief delayed ,stranded communities
Households Missing aid	40,500 people needed aid(Sri Lanka , 2024)	Delayed support , health issues, loss of livelihood
Poor Warnings	30% Globally not covered by early warning system	people unprepared , higher morality/ damage
Unsafe habits/health	45,448 dengue cases in 2024(Sri Lanka)	Disease outbreaks, deaths, especially children

Target Audience



Residents

Receive timely alerts, safe evacuation routes, and awareness tips.



**Local
authorities/emergency
services**

Can coordinate evacuations efficiently and respond faster to emergencies.



**Aid/relief
organizations**

Can distribute aid accurately, reaching all households, including hidden or remote areas.

Aim and Objectives



Awareness & Education

Visualizations & real-time risk alerts.



Evacuation Support

Community reporting & safe route guidance



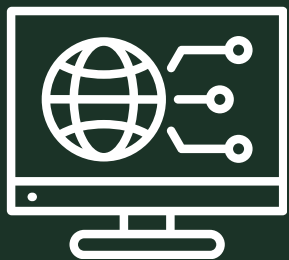
Health & Safety Guidance

Safety checklists & post-disaster tips



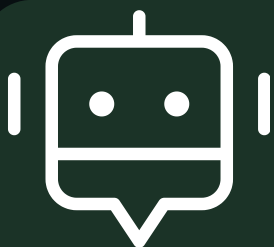
Fair Relief Distribution

Pre-registered households ensure everyone receives aid



Technology Integration

Maps, weather data, and SMS alerts for all communities



Future Readiness:

AI predictions & multi-language / low-connectivity support.

Proposed Solution (Overview)



Proposed Solution (Overview)

A disaster preparedness web app that unites communities, authorities, and aid providers to improve awareness, evacuation, relief, and recovery.

One platform – many stakeholders – safer communities.

01

Visualization of Risks



Maps, color-coded alerts, predictive updates.

02

Community Alerts



Real-time reports & safe route guidance.

03

Pre-Registration



Household data for fair and efficient aid distribution.

04

Health & Safety Guidelines



Tailored checklists & tips for disaster survival.

05

Multi-Stakeholder Access



Separate dashboards for residents, local authorities, and NGOs, ensuring coordinated disaster response.

**What Makes Our App
Stand Out**

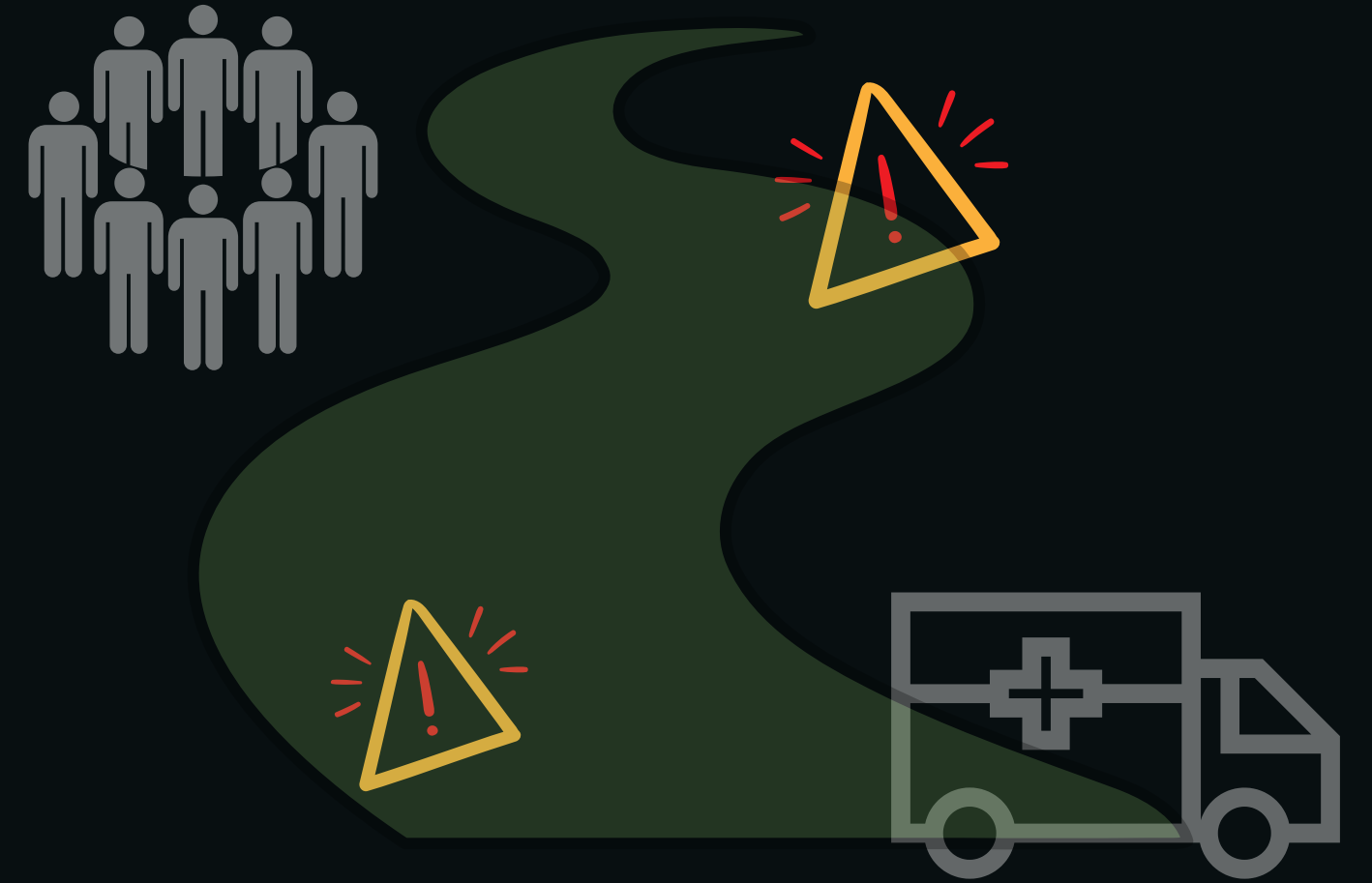


Visual representation of Risks

Maps showing disaster-prone areas

Color-coded alerts: Green = Safe, Yellow = Warning, Red = High Risk

Predictive updates (e.g., if it rains, risk increases)



Community alerts and safe route guidance

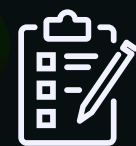
Crowdsourced reporting of blocked roads / hazards

Alternative safe route suggestions

Reduces traffic congestion and helps emergency responders



Pre-Registration for **Disaster Zones**



Household form (members, contact, special needs)



Data shared with authorities for relief & evacuation



Offline/Community registration (via GN officers)

Multi-Stakeholder Access





Health and Safety Guidelines

Preventing casualties during and after the disaster



Tailored checklists for different disasters



Critical health advice to prevent disease



Emergency preparedness and reminders



post disaster recovery guidance

Technology Behind Our Web App

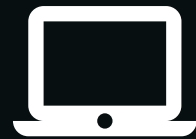
Technology Behind Our Web App

Platform (User)



- Web-based application (accessible via PC & mobile browsers)
- Responsive design ensures usability across devices

Frontend



- Interactive maps & dashboards
- Color-coded alerts, notifications, and visual risk representation
- Technologies (examples): React / Angular / Vue.js

Backend



- Handles predictive analytics using historical disaster & rainfall data
- Manages user accounts, household registration, and alerts
- Technologies (examples): Node.js / Express / Firebase
- Predictive analytics (rainfall+history)

Integration



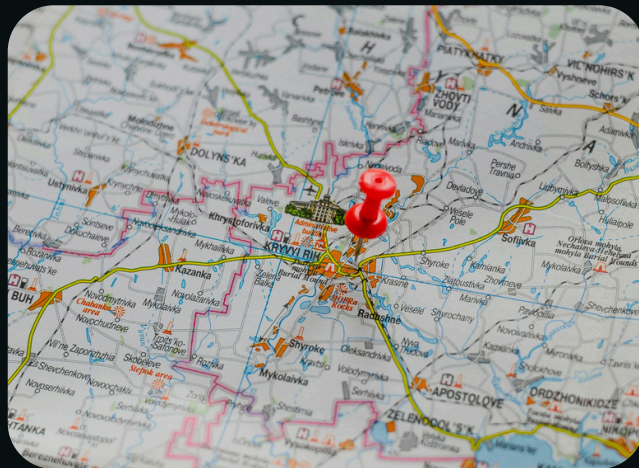
- Google Maps API for maps & route suggestions
- Government disaster alert feeds
- SMS / Email notifications for areas with low connectivity

Deployment



- Host on a cloud platform (AWS / Google Cloud / Azure)
- Ensures scalability (can handle sudden traffic during disasters)

Innovation



Visual Awareness

Color coded maps to show disaster risks clearly



Community Reporting

- Real-time, crowdsourced reports from citizens on the ground.
- Provides live updates



Fair relief distribution

Pre-registration system ensures no one is missed during aid distribution



Health & Safety Preparedness

Provides preparedness checklists and tips

Feasibility And Market plan



Starts Pilot in high-risk districts



Collaborate with Disaster Management Centre (DMC), NGOs, and local authorities



sustainable model with Government adoption or NGO sponsorship.



Low cost High impact

Impact



SAVES LIVES

Increased awareness leads to earlier, more effective evacuations.



Ensures Equity in Aid

Pre-registration guarantees fair distribution of relief supplies



REDUCES CHAOS

Faster evacuations with safe route guidance and prevents congestion and panic



BETTER HEALTH RECOVERY

Post-disaster guidelines prevent disease and secondary casualties

Current situation

Confusion & Ignorance:

People ignore text warnings because they don't understand the risk

Chaotic Evacuation:

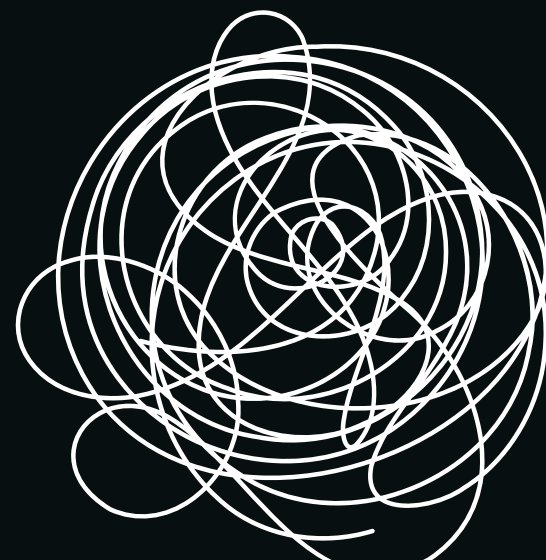
Everyone uses the same blocked roads, causing more traffic jams

Unfair Aid:

Hidden communities are missed; and aid distribution is inefficient

Post-Disaster Illness:

Lack of knowledge leads to waterborne diseases and health crises



Our impact

Informed & Prepared:

People see clear color-coded maps and understand the risk

Orderly Evacuation:

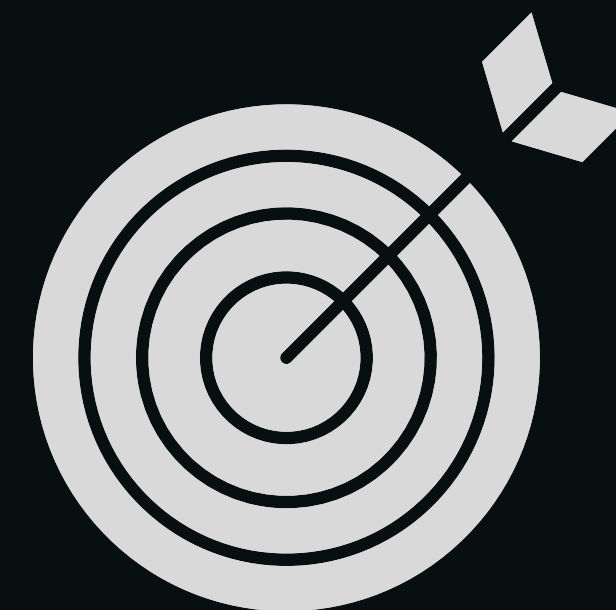
Live community reports suggest safe, alternative routes

Fair Aid:

Pre-registration ensures every household is accounted for and reached

Healthier Recovery:

Tailored guidelines prevent outbreaks and save lives after the disaster



SCALABILITY AND FUTURE ENHANCEMENTS



Advanced Prediction:

Integrate AI/ML for more accurate disaster forecasting using rainfall and soil data



Greater Accessibility:

Multi-language support (Sinhala, Tamil, English)

Offline SMS/USSD alerts for areas with poor internet connectivity



Expanded Features:

Integration with river water levels & satellite imagery for stronger disaster predictions

Conclusion

Our solution prepares communities, saves lives, and ensures fair disaster response. By combining awareness, safe evacuation guidance, and equitable aid distribution, we can transform disaster management in Sri Lanka.

Join us to build a safer, more resilient Sri Lanka for everyone.

PREPARE.PROTECT.PREVAIL



THANK YOU
